SAMEEK BHATTACHARYA

sameek.official@gmail.com | LinkedIn | Website | GitHub | +1 940 597 2285

PROFESSIONAL SUMMARY

AI/ML and **Software Development** professional with more than 4 years of experience across software development, research and deployment, skilled in Python, PyTorch, Scikit-learn and deep learning models among others. Ability to architect end-to-end pipelines in adversarial audio attacks, RAG systems, NLP and computer vision, translating research into real-world applications. Actively seeking AI/ML Engineer, Data Scientist and Software Developer roles, leveraging an MS in Artificial Intelligence from the University of North Texas.

WORK EXPERIENCE

University of North Texas — *United States*

Data Scientist Intern — *May 2025 – July 2025*

- Developed a student progress monitoring system using **LLMs** and **NLP techniques** and set up a dashboard using **Tableau and Seaborn** to show live progress reports which increased efficiency in **monitoring workflow by 20%** and to track student performance. Hosted the application in **AWS Endpoint** and connecting to **AWS S3** and **RDS**.
  - Setup Sentiment Analysis, Language Embedding, and Question answering model using **PyTorch**. Experienced with **BERT**, **RoBERTa** models and **TF-IDF NLP** implementation. Developed backend code using **Python** and experienced with data handling using **Pandas** and **Dask**, and handling data formats like **CSV, Parquet** etc.
  - Experienced in setting up the **ollama** library to deploy LLM models locally. Experienced in using **Llama** for **LLM applications** and **BGE** model for **generating embeddings** which is used for student report generation.
  - Experienced in hosting the dashboard app in **Streamlit**. Also used **Gradio** for **frontend development** and **ngrok** for **hosting webpages**.

Research Assistant (Visual Computing and Biometric Security Lab) — *Feb 2025 – Current*

- Developed a **GAN-based adversarial attack** for audio data and evaluated against existing state-of-the-art models, outperforming other white box attack mechanisms. Achieved **Attack Success Rate 99.11%+**, **attack speed 0.007 sec**, making it possible to be implemented in real-time audio systems. Submitted to **Interspeech 2026**.
  - **Datasets, Tasks & Models: ASVspoof2019** – Deepfake detection (**Xception**, **ResNet50 feature extractor** with **Logistic Regression classifier**), **DCASE2019** and **UrbanSound8k** – Sound event classification (**PANN-CNN14, ResNet50**), **LibriSpeech** – Speaker Verification (**ECAPA-TDNN** with **cosine similarity**)
  - **Adversarial Attacks: Gaussian Noise**, **FGSM**, **BIM**, **PGD**, **DeepFool**, **Carlini Wagner** and our **GAN-based Attack**.
  - **Architectural Pipeline:** Extracted **MFCC** and **STFT** features and saving **NumPy** arrays using **librosa** library from raw **.flac** and **.wav** data. Passed through models for training and evaluation. Implemented attacks and computed attack evaluation. Extracted audio embeddings for speaker verification. Computed cosine similarity scores.

Teaching Assistant (Spring 2026: Introduction to AI | Fall 2025: Computer Science I) — *Aug 2025 – May 2026*

- Assisted in developing **course materials** and **led lab sessions** for both courses – **90+ students** in each class, mentoring students on **C/C++ programming**, **search algorithms** and **fundamental AI concepts**. Processed student data using **SQL** and Graded assignments with **constructive feedback**.

Institute of Engineering and Management — *India*

Research Assistant — *Jan 2023 – May 2024*

- Developed **AI-driven tools** including a **life-cycle management framework** with **result dashboard** and a **benchmarking system** for **tabular and image datasets** using **Tableau**, **Scikit-learn** and **TensorFlow**.
- Analysed **student performance data** to provide actionable insights to the **education board**.

Tata Consultancy Services — *India*

System Engineer — *Aug 2021 – Nov 2022*

- Developed and maintained **Johnson & Johnson's training website** for doctors and staff, working as a **full-stack web developer** using **HTML**, **CSS**, **Bootstrap**, **JavaScript**, **Git**, and **Atlassian tools** and **Drupal** as the CMS following agile framework.
- Ensured **Johnson & Johnson compliance standards** through certifications in **iQMS**, **HSE**, **Data Privacy**, and **Health & Safety**.

EDUCATION

University of North Texas (TX, United States) — **July 2026**
*Masters (Thesis) – Artificial Intelligence – **MS Distinguished Scholar*** — *GPA – 3.87/4.0*

University of Engineering and Management (WB, India) — **June 2021**
Bachelors – Computer Science and Engineering — *GPA – 8.59/10.0*

PUBLICATIONS (P: PUBLISHED, A: ACCEPTED)

A: Exploiting Neural Audio Codec Latents for Adversarial Audio Attacks. (Interspeech 2026)

P: Analysis of DNA Cryptography's Improvements for the Modeling of Secured Corpus. (ICCASA 2023)

P: Finding a baseline machine learning model in Sci-Kit Learn package in terms of accuracy and efficiency. (IEEE Pune Section International Conference 2021)

PROJECT EXPERIENCE

- **RAG-Powered Interactive Document QA System** *(Google Gemini API, gemini-2.5-flash, google-genai, python-dotenv, Python)*
  - Built a dual-mode QA pipeline a simple Q&A flow routing queries directly to the **Gemini API**, and a **RAG** mode that prepends user-supplied document context to the prompt, with both modes maintaining full conversation history and generating a **session summary** on exit; the system handles requests sequentially with **structured error handling**.
  - Achieved **context-aware**, **hallucination-reduced** responses grounded in user-provided documents, with an average **response time** of **5–15 seconds**, and a default **temperature** of **0.7** balancing creativity with accuracy across use cases such as document Q&A, CV-to-job-description matching, and code review.
- **EasyChords – Application for Playing Piano Chords** *(Python, Eel, Pygame, NumPy, Pillow, PyInstaller, HTML/CSS/JS, GitHub Copilot)*
  - Architected a hybrid desktop app with a **Python** backend connected to an **HTML-CSS-JS** frontend via **Eel**; the audio engine performs **MIDI-to-frequency** conversion, builds major/minor scales across seven harmonic degrees, and synthesizes multi-harmonic sine waves with a custom ADSR envelope via **NumPy**, while **Pygame** handles **low-latency** chord playback.
  - Delivered a fully **zero-dependency**, **offline Windows** executable packaged via **PyInstaller** with an embedded custom circular icon, featuring an adjustable sustain control for post-release chord fade-out.
- **Self-Sustaining Wearable System for Solar Panel-Based UV Exposure Monitoring** *(ESP32, INA219, UV Sensor, Solar Panel, Arduino C++, Python, scikit-learn, Jupyter)*
  - Collected ~**5,000** simultaneous readings from an **ESP32**-based wearable integrating an **INA219** power monitor, **UV sensor**, and **solar panel**; data was normalised, visualised, and split **70/15/15** for train/validation/test, then fed into **Logistic Regression** and **SVM** classifiers.
  - **Logistic Regression** achieved **97% accuracy**, outperforming **SVM** (**96%**) and a **baseline model** (**93%**); solar panel output validated as a **cost-effective**, **self-powering** proxy for UV intensity.
- **Lightweight Automated Essay Scoring (AES) via Regression and Optimized Feature Selection** *(Python, scikit-learn, pandas, NumPy, TF-IDF, PCA, Jupyter Notebook)*
  - Built an AES pipeline on **141 essays**; extracted **1,221 raw TF-IDF features**, reduced to 53 via **Univariate Feature Selection**, applied **PCA**, then trained and compared regression and classification models for ordinal score prediction.
  - **Gradient Boosting Regressor** achieved **QWK 0.9595** and **RMSE 0.9264**, significantly outperforming **KNN** (**QWK 0.6266**).
- **Reddit Comment Sentiment Classification: TF-IDF Baseline vs. Fine-Tuned RoBERTa** *(Python, HuggingFace Transformers, RoBERTa, TF-IDF, scikit-learn, Kaggle)*
  - Subsampled **30,000** comments from the **Kaggle 1M Reddit** dataset; applied **TF-IDF** for a **Logistic Regression** baseline and fine-tuned **RoBERTa** for contextual sentiment classification across positive, neutral, and negative classes.
  - Fine-tuned **RoBERTa** outperformed the baseline with **F1-scores** of **0.65 (positive)** and **0.47 (neutral)**; the baseline degraded sharply under class imbalance, confirming transformer contextual embeddings are critical for noisy social media data.
- **CNN Architecture Comparison for Binary Skin Cancer Classification on ISIC Dataset** *(Python, TensorFlow, Keras, ResNet50, DenseNet121, VGG16, InceptionV3, OpenCV)*
  - Comparative study of **ResNet50, DenseNet121, VGG16**, and **InceptionV3** on **ISIC dermoscopic images** for binary benign/malignant classification, evaluating **CLAHE** contrast enhancement and polynomial feature transformation.
  - **DenseNet121 + CLAHE** achieved the best result with **F1 0.89** for both classes, outperforming all other architectures.

SKILLS

Programming Languages: Python, Java, C, HTML, CSS, C++, JavaScript, SQL
Data Science & ML: Retrieval-Augmented Generation (RAG), Audio Processing, Deepfake detection, Sound event classification, Image Processing, Computer Vision, Feature Engineering, LLM, Natural Language Processing, Deep Learning (ECAPA, CNNs, ResNet, Xception), traditional ML models, multimodal data processing, Scikit-learn, Transfer learning, Model fine-tuning
Cloud Technologies: Kaggle, GoogleWorkspace(Colab, Slides), GitHub, Atlassian(Bitbucket), Linux, AWS (RDS, S3, Endpoint)
DevOps & Version Control: Anaconda, Git, CI/CD
Frameworks & Libraries: PyTorch, TensorFlow/Keras, Hugging Face, Weights & Biases, Gemini-API, Streamlit, Dask, Bootstrap, Jupyter Notebook, VS Code, Atlassian (Jira, Confluence), Regression Testing, A/B Testing, Agile Framework, Copilot, ChatGPT
Visualization Tools: Tableau, Matplotlib, Seaborn, Plotly
Other: LeetCode, Arduino, Excel, Adobe Illustrator, Adobe Effects, InShot

COMMUNITY SERVICE EXPERIENCE

Kolkata Robotic Society — *India*
Co-Founder — *Sept 2017 – May 2020*

- Co-founded a **non-profit educational organization** to promote **tech awareness** by teaching **high school students AI**, **machine learning**, **robotics**, **Arduino**, and **drones**, reaching **150+ participants** through approx. **20 sessions** and a **state-level workshop**.
- Developed **entrepreneurial skills** by engaging with **investors** and **entrepreneurs** to support and grow the organization.